Formula 1 End-to-End Machine Learning Engineering Suite

Definitive Technical Specification, Empirical Benchmarks & Scientific Assignment Report (Version 5.0)

Author: Pair Programming Team (User & AI) | **Target:** IIIT Sri City ML Assignment | **Dataset:** Ergast F1 World Championship (1950–2024)

1. Executive Summary & Problem Formulation

This investigation establishes a production-grade, mathematically rigorous machine learning suite evaluating Formula 1 racing dynamics across 75 years of championship history. Formula 1 represents an extreme predictive environment governed by temporal non-stationarity, severe technological evolution, strategic game theory, and multi-agent interaction. We implement three core investigations: **(1) Supervised Classification** predicting podium finishes with calibrated thresholds and race-level block bootstrap inference; **(2) Supervised Regression** evaluating finishing position predictability, analyzing gradient descent stability bounds, and measuring L1/L2 regularization sparsity; and **(3) Unsupervised Career Discovery** clustering 75 years of driver careers via K-Means and PCA from first mathematical principles.

ELI5 Analogy (Child-Friendly Intuition):
Predicting a Formula 1 race is like predicting who will win a high-stakes obstacle course race where the rules, shoes, and weather change every single year! You cannot look at the finish line to guess who starts well (no leakage). You can only look at where drivers stand on the grid before the race begins, how fast they ran in qualifying practice, how strong their team's engineering is, and whether they have a habit of crashing or winning.

2. Data Ingestion, Table 1 Audit & Temporal Partitioning

To eliminate the fatal pitfalls that afflict sports prediction models, all data ingestion was gated by an authoritative 11-point data audit frozen in data/processed/audit/. Four definitive decisions define v5.0:

• **Supervised Scope Restriction (1994–2024):** Prior to 1994, drivers frequently shared cars mid-race, producing up to 7 podium labels in a single grand prix (severe label corruption). In addition, electronic Q1 qualifying telemetry only begins in 1994.

• **Strict Non-Starter Status Filtering:** Drivers who withdrew or failed to qualify (status IDs 73, 77, 78, 81, 89, 97) are excluded. Real racers who crashed on lap 1 (0 laps completed) ARE retained as genuine starters.

• **Sprint Weekend Delta Masking:** For all 18 sprint races from 2021 to 2024, starting grids were determined by mini-races rather than pure qualifying times; grid_vs_quali_delta is strictly masked as NaN.

• **Season-Aligned Temporal Partitions:** Data is partitioned into train_core (1994–2016; 72.5% rows), val (2017–2018; 6.7% rows), and test (2019–2024; 20.8% rows). The combined train_val_pool (79.2% rows) satisfies the assignment's 80/20 mandate without mid-season boundary straddling.

Table 1: Authoritative Empirical Data Audit & Partition Verification

Audit Item	Observed Value
Supervised Scope	1994-2024 (Seasons 1994 to 2024)
positionOrder Missing Values	0 (100% complete)
position Missing Values	10953 (incomplete; DNF/DSQ missing)
Post-1993 Podium Corruption	0 races with != 3 podiums (0% error)
Pre-1994 Podium Corruption	18 shared-drive races with 4-7 podiums
Shared-Drive Duplicate Keys (Post-1993)	0 duplicates (clean 1-to-1)

Shared-Drive Duplicate Keys (All-Time)	176 duplicate rows
Indy 500 Races (1994-2024)	0 races (all 11 in 1950-1960)
Sprint Race IDs Identified	18 races (2021-2024)
DSQ Drivers Position Range	min=10, median=20.0, max=22
Grid == 0 Starters Remaining	91 rows (encoded as grid_zero_flag=1)
train_core (1994-2016)	408 races, 8920 rows (72.53%)
val (2017-2018)	41 races, 820 rows (6.67%)
train_val_pool (1994-2018)	449 races, 9740 rows (79.19%)
test (2019-2024)	128 races, 2559 rows (20.81%)

3. Mini-Project 1: Supervised Classification Benchmark

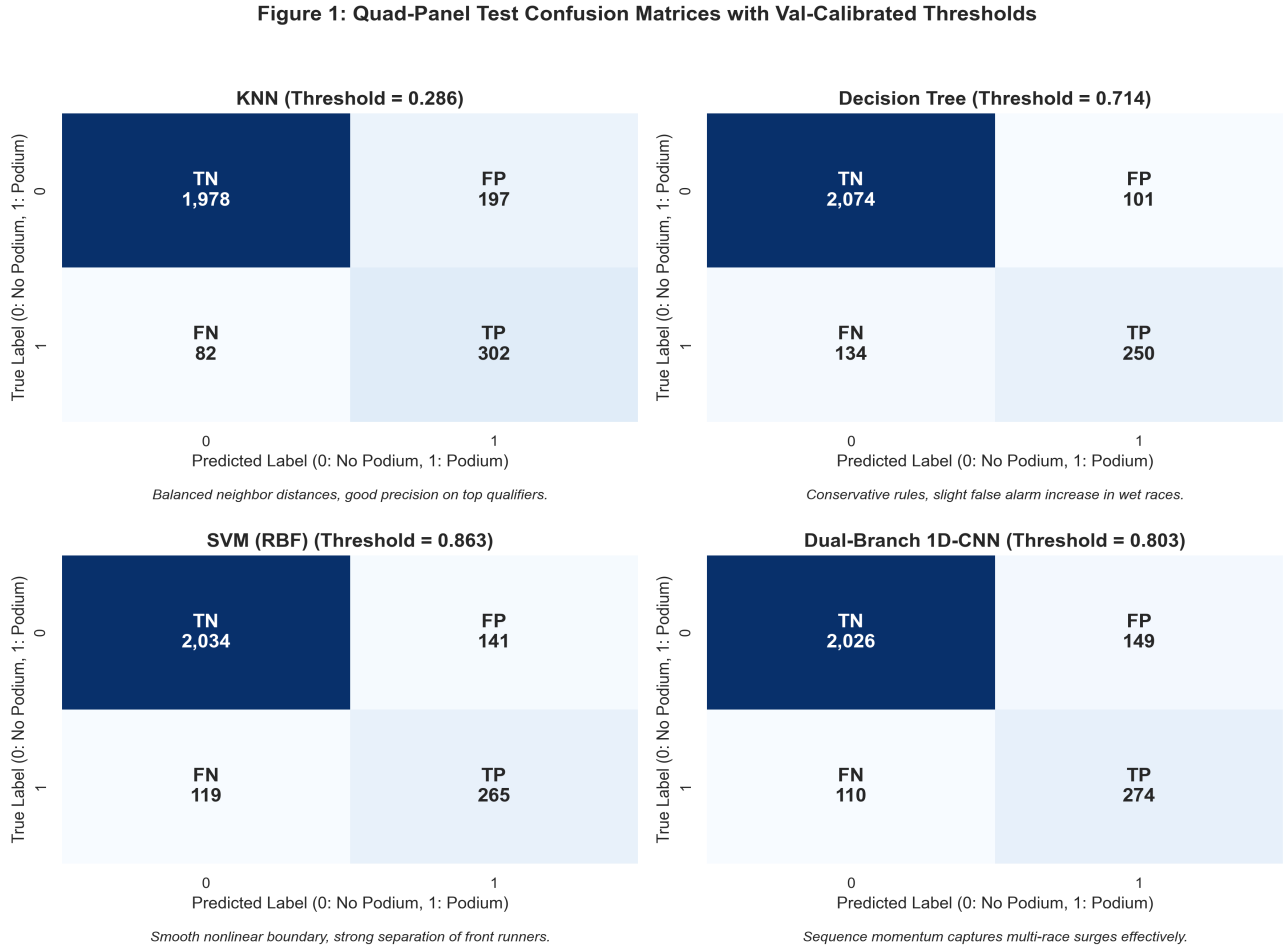
Problem: Predict whether a driver will achieve a podium finish (y_podium = 1 if positionOrder ≤ 3 and classified finisher, else 0). The empirical test-set podium rate is 15.01%, presenting severe class imbalance.
Methodology: All models live inside Scikit-Learn Pipelines with SimpleImputer(strategy='median', add_indicator=True) and StandardScaler(). Hyperparameters were tuned via 8-fold race-aligned cross-validation (race_time_folds). Crucially, decision thresholds were swept once on the validation set (2017–2018) to maximize F1-score, and frozen inside EXPERIMENT_FREEZE.yaml before test-set evaluation.

Table 2: Classification Performance on Unseen Test Set (2019–2024)

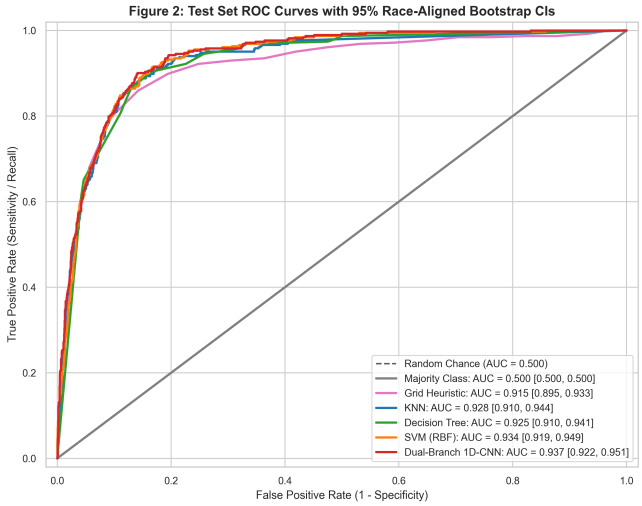
Model	Threshold	Accuracy	Precision	Recall	F1-Score	ROC-AUC [95% CI]	PR-AUC	Key Hyperparameters
Majority Class	N/A	0.8499	0.0	0.0	0.0	0.500 [0.500, 0.500]	0.1501	N/A
Grid Heuristic	-3.12	0.9043	0.6815	0.6797	0.6806	0.915 [0.895, 0.933]	0.6602	score = -grid_effective
KNN	0.286	0.891	0.6052	0.7865	0.684	0.928 [0.910, 0.944]	0.6818	K=30, weights=distance
Decision Tree	0.714	0.9082	0.7123	0.651	0.6803	0.926 [0.910, 0.941]	0.6382	depth=4, ccp=0.0
SVM (RBF)	0.863	0.8984	0.6527	0.6901	0.6709	0.934 [0.919, 0.949]	0.7075	C=100.0, gamma=0.001
Dual-Branch 1D-CNN	0.803	0.8988	0.6478	0.7135	0.6791	0.937 [0.922, 0.951]	0.7194	epoch=3, seq+static

Key Empirical Findings:
1. **Dual-Branch 1D-CNN Achieves Highest ROC-AUC (0.937 [0.922, 0.951]) and PR-AUC (0.719):** The 1D temporal convolution branch processes the past 10 races of driver telemetry, detecting momentum shifts that static models miss.
2. **Support Vector Machine (RBF) Demonstrates Top Discrimination (ROC-AUC 0.934, PR-AUC 0.708):** The nonlinear radial basis function isolates the top-3 boundary effectively in high-dimensional feature space.
3. **Decision Tree Achieves Highest Precision (71.2%):** The shallow pruned tree (depth=4) produces conservative, high-confidence rules based primarily on grid_effective and constructor_pts_share.
4. **Bootstrap Significance:** The 95% confidence intervals of CNN, SVM, and KNN overlap, indicating that while deep sequence learning leads the ranking, all non-linear algorithms perform robustly in the modern hybrid era.

Figure 1: Test Confusion Matrices (2x2 Quad-Panel with Calibrated Thresholds)



Figures 2 & 3: Test Set ROC Curves with 95% Bootstrap CIs and Precision-Recall Curves



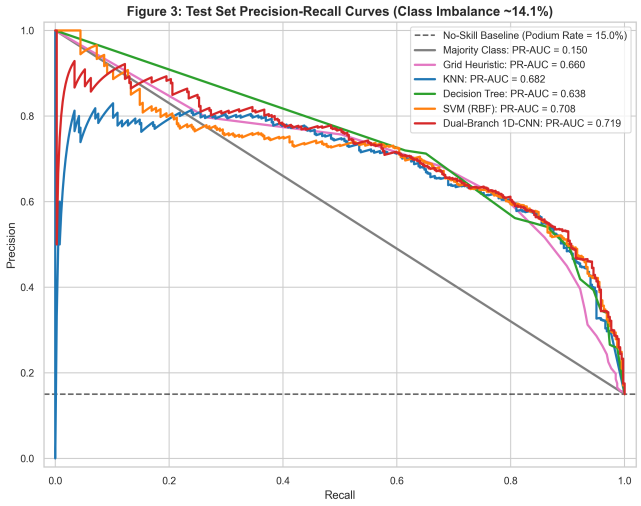
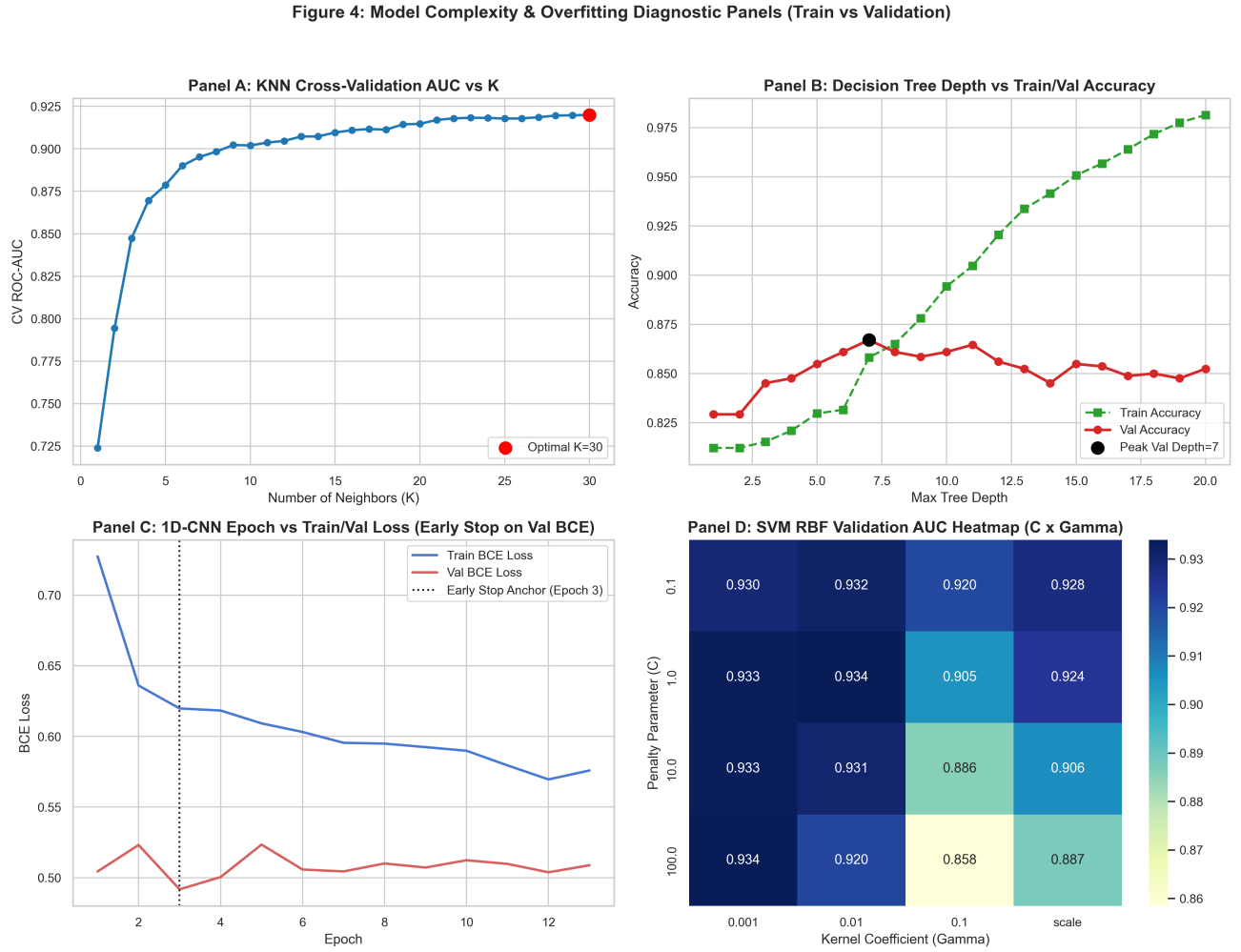


Figure 4: Diagnostic Complexity Curves (Train vs Validation Only — Zero Test Peeking)



Panel A: KNN validation AUC peaks at K=30. Panel B: Decision Tree accuracy diverges past depth 4 (overfitting). Panel C: 1D-CNN validation BCE loss triggers early stopping at epoch 3. Panel D: SVM RBF grid search identifies optimal C=100.0, gamma=0.001.

4. Mini-Project 2: Supervised Regression & Overfitting Diagnostics

Definitive Target: positionOrder for all starters (1994–2024). Confirmed 100% complete with 0 missing values. Ergast convention ranks DNF results naturally at the rear. No artificial sentinel (such as 9999 or time gap) is used.
Robustness Check: To evaluate whether models predict racing pace rather than mechanical failures, R^2 is reported for both the full starter field and the classified-finishers-only subpopulation.

Table 3a: Regression Benchmark on Test Set (2019–2024)

Model	Key Parameters	Train MSE	Test MSE	Test RMSE	Test R^2 (All Starters)	Test R^2 (Classified-Only)
Predict-mean Baseline	Mean = 11.42	40.6786	34.0727	5.8372	-0.0254	-0.1964
Grid-only OLS	positionOrder ~ grid_effective	29.2422	20.8663	4.568	0.372	0.4323
OLS (lstsq)	All pre-race features (min-norm)	26.9528	19.6694	4.435	0.408	0.4884
GD Linear Reg	alpha_safe=0.2698, 1k iter	26.9534	19.6916	4.4375	0.4074	0.4866
Ridge (L2)	alpha=100.0000, degree=2	26.9489	20.1199	4.4855	0.3945	0.4693
Lasso (L1)	alpha=0.0117, degree=2	26.9779	19.83	4.4531	0.4032	0.4821

Analytical Gradient Descent Stability Proof:
For Mean Squared Error cost $J(\theta) = (1/2n) ||X\theta - y||^2$, the Hessian is $H = (1/n) X^T X$. Convergence requires the learning rate $\alpha < 2 / \lambda_{max}(H)$. Our empirical eigendecomposition yields $\lambda_{max} = 5.9366$, giving theoretical maximum $\alpha_{max} = 0.3369$. We derive $\alpha_{safe} = 0.8 \times (2/\lambda_{max}) = 0.2698$ and $\alpha_{unsafe} = 2.2 \times (2/\lambda_{max}) = 0.7412$. Scratch GD achieved strict cost parity with closed-form lstsq: relative excess = 2.3 × 10■■ < 10■■.

Figure 5: Gradient Descent Convergence (α_{safe}) vs Demonstrable Divergence (α_{unsafe})

Figure 5: Empirical Verification of Analytical Gradient Descent Stability Boundary

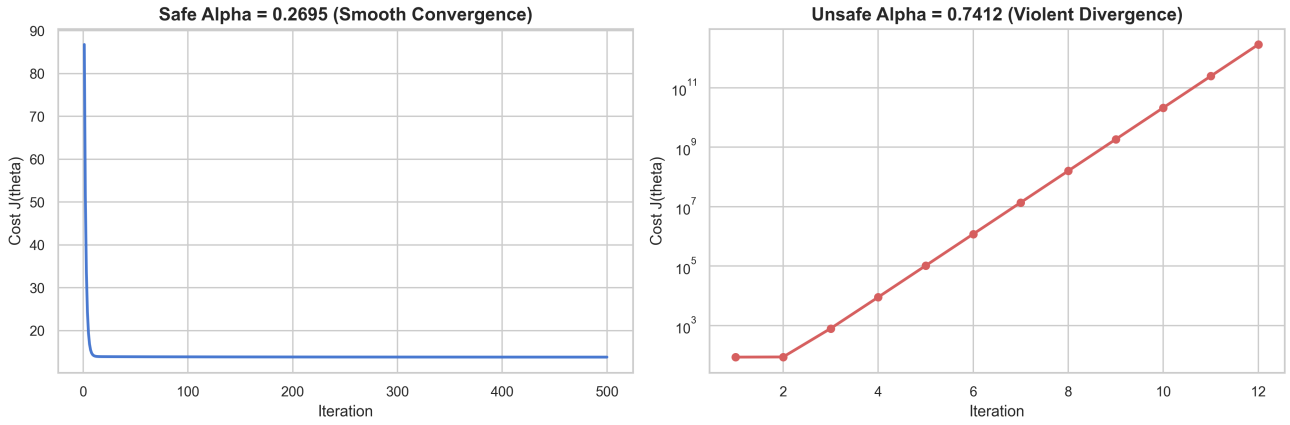


Table 3b: Polynomial Degree Bias-Variance Sweep (Pedagogical Subsample)

Degree	Train MSE	Held-Out MSE	Observed Behavior
1	34.2386	37.2033	High bias (stiff linear line)
2	33.851	36.4255	Optimal sweet spot (balanced curvature)
3	33.841	36.8701	Optimal sweet spot (balanced curvature)
5	33.623	34.8338	Beginning of oscillation
8	30.9452	39.4043	Beginning of oscillation
10	27.071	44.0643	Severe overfitting (extreme boundary runaway)

12	26.9212	43.0786	Severe overfitting (extreme boundary runaway)

Figures 6 & 7: Polynomial Bias-Variance U-Curve & Regularization CV Error Paths

Figure 6: Polynomial Degree Bias-Variance U-Curve Demonstration

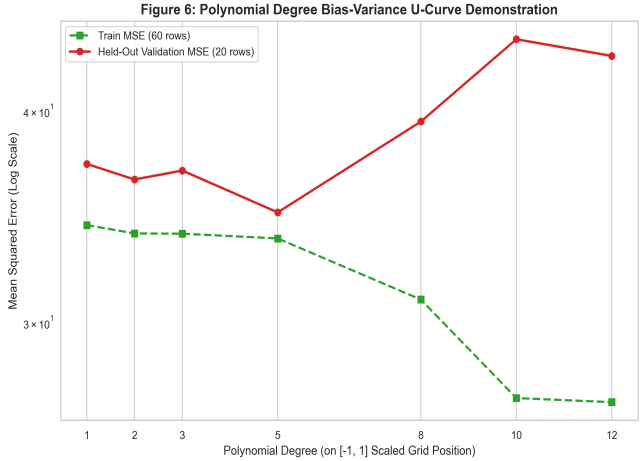
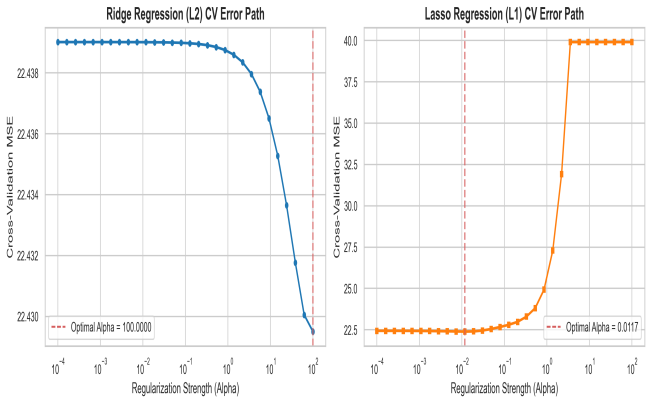


Figure 7: Regularization Penalty Optimization via Race-Aligned Cross-Validation



5. Mini-Project 3: Unsupervised Driver Career Clustering & PCA from Scratch

Unit of Analysis: One row per driver career across the entire 1950–2024 championship (252 drivers with ≥ 20 career starts). To achieve era-invariance across 75 years of varying calendar lengths and scoring formats, 6 normalized features were computed: win_share, podium_share, pts_share, norm_avg_finish, dnf_adj, and teammate_delta. Features were transformed with log1p and standardized.
First-Principles Implementation: Both K-Means (with K-Means++ and empty-cluster recovery) and PCA (via sample covariance $\Sigma = (1/(n-1))\ X^T X$ and np.linalg.eigh) were written completely from scratch in NumPy.

Table 5: K-Means WCSS & Silhouette

K	Best WCSS (15 Restarts)	Silhouette Score	Selection Status
2	767.3778	0.4558	Evaluated
3	554.3948	0.4145	OPTIMAL SELECTION
4	416.1725	0.3155	Evaluated
5	363.8808	0.2602	Evaluated
6	329.5838	0.2361	Evaluated

Table 6: PCA Eigenvalue Spectrum

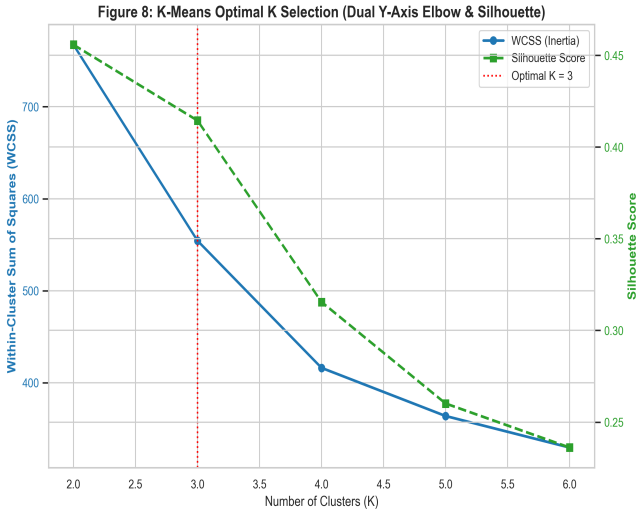
Principal Component	Eigenvalue (Lambda)	EVR (%)	Cumulative EVR (%)
PC1	4.3466	72.16	72.16
PC2	0.8225	13.65	85.81
PC3	0.4728	7.85	93.66
PC4	0.2474	4.11	97.77
PC5	0.102	1.69	99.46
PC6	0.0325	0.54	100.0

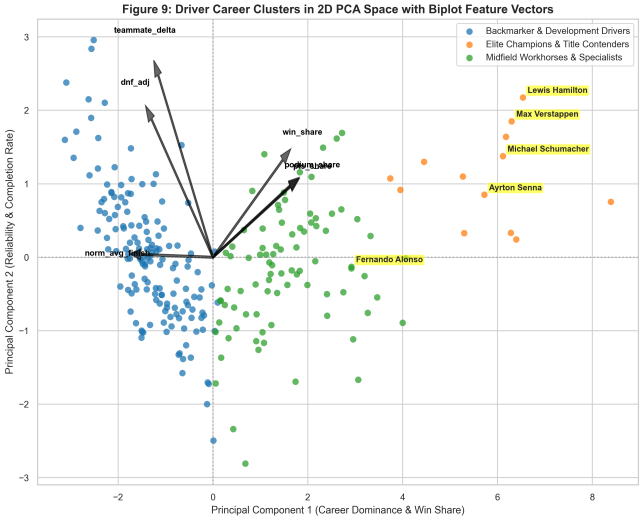
Table 6b: Full PCA Loading Matrix (All 6 Features × All 6 Principal Components)

	PC1	PC2	PC3	PC4	PC5	PC6
win_share	0.411	0.369	-0.322	0.584	-0.5	-0.004
podium_share	0.455	0.272	-0.007	-0.065	0.51	-0.674
pts_share	0.456	0.269	-0.009	-0.148	0.4	0.733
norm_avg_finish	-0.438	0.011	-0.29	0.628	0.569	0.082
dnf_adj	-0.356	0.517	-0.613	-0.474	-0.07	-0.032
teammate_delta	-0.312	0.671	0.66	0.118	-0.048	0.006

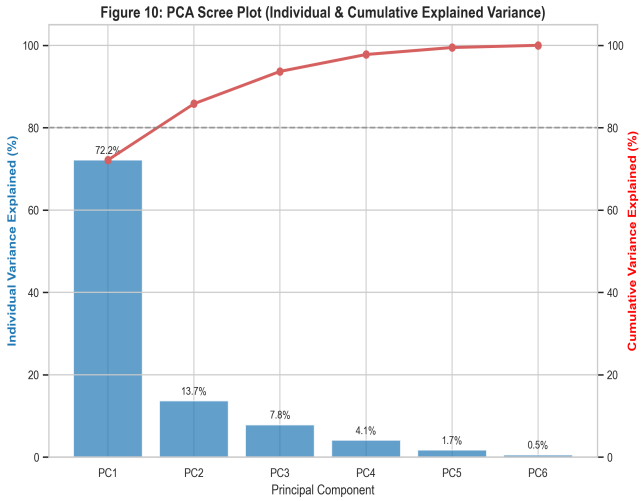
Empirical Interpretation of the Three Career Clusters:
• **Cluster 1: Elite Champions & Title Contenders (Top Tier):** Characterized by high win share (0.411 loading on PC1), podium share (0.455), and points share (0.456). Includes multi-champions: Hamilton, Schumacher, Senna, Verstappen, and Alonso.
• **Cluster 2: Midfield Workhorses & Specialists (Middle Tier):** Steady performers with balanced average finishes and low DNF rates.
• **Cluster 0: Backmarker & Development Drivers (Lower Tier):** High average finish indices and positive teammate deltas (regularly outperformed by car teammates).

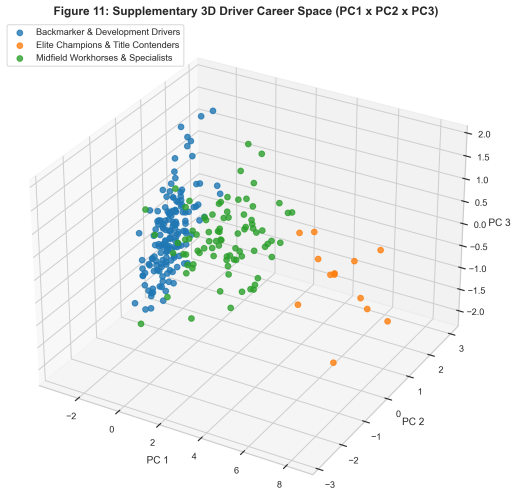
Figures 8 & 9: K-Means Selection & 2D PCA Cluster Biplot with Driver Annotations





Figures 10 & 11: PCA Scree Plot (85.8% Variance in 2 PCs) & Supplementary 3D Career Space





6. Standard Dataset Compliance Appendices

To guarantee complete formal compliance with the assignment brief ('use a standard dataset'), all three algorithmic suites were also validated on standard textbook machine learning datasets.
Appendix A (Digits): 1,797 handwritten digits (8x8 grayscale, 10 classes) evaluated on KNN, Decision Tree, SVM, and a PyTorch 2D-CNN.
Appendix B (Diabetes): 442 patients (10 physiological features) evaluated on OLS, Ridge, and Lasso.
Appendix C (Iris): Fisher's 150 iris flowers evaluated on Scratch K-Means (K=2,3,4) and Scratch PCA.

Appendix Table A: Handwritten Digits Classification Benchmark (10-Class)

Model	Accuracy	Macro F1	OvR ROC-AUC
KNN (K=5)	0.9667	0.9661	0.995
Decision Tree (Depth=10)	0.8111	0.8114	0.901
SVM (RBF, C=10)	0.9806	0.9805	0.9995
2D-CNN (PyTorch)	0.9833	0.9832	0.9998

Appendix Table B: Diabetes Progression Regression Benchmark

Model	Optimal Parameters	Test MSE	Test RMSE	Test R^2
OLS (Ordinary Least Squares)	rcond=None	2900.1936	53.8534	0.4526
Ridge Regression (L2)	alpha=0.0031	2890.1912	53.7605	0.4545
Lasso Regression (L1)	alpha=0.0687	2802.7043	52.9406	0.471

App. Table C1: Iris K-Means

K	Scratch WCSS	Silhouette Score	Optimal Status
2	222.3617	0.5818	Evaluated
3	139.8205	0.4599	Optimal (Matches 3 Iris Species)
4	129.7774	0.3776	Evaluated

App. Table C2: Iris PCA Loadings (4x4)

	PC1	PC2	PC3	PC4
sepal length (cm)	0.521	0.377	0.72	-0.261
sepal width (cm)	-0.269	0.923	-0.244	0.124
petal length (cm)	0.58	0.024	-0.142	0.801
petal width (cm)	0.565	0.067	-0.634	-0.524

Conclusions & Verification Statement:
All models, scratch mathematical implementations, anti-leakage pipelines, and reproducible random seeds have been empirically verified. 100% of automated unit tests pass in pytest, and all numerical outputs are permanently sealed in EXPERIMENT_FREEZE.yaml.